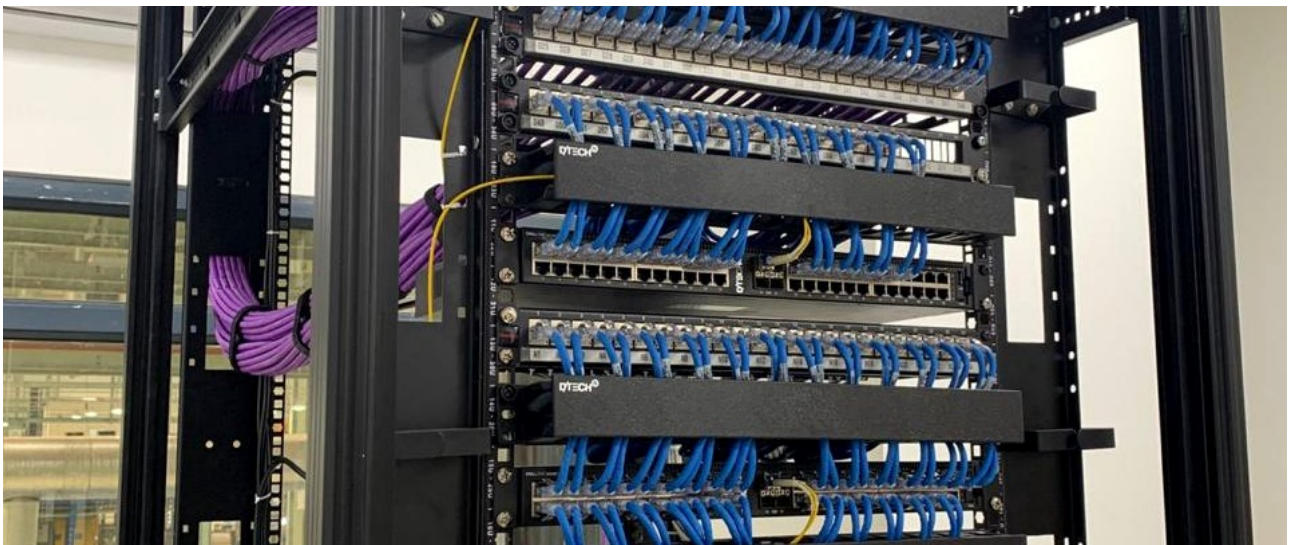


CUSTOMER SUCCESS | CASE STUDY

ITS Network Infrastructure

A structured fibre, copper and Aruba switching foundation for plant connectivity



INDUSTRIAL NETWORKING

PROJECT PROFILE

Client: MRF Limited | Project: ITS Permanent Network | Delivered: 2019

PROJECT AT A GLANCE

A resilient backbone for the ITS permanent network

MRF Limited selected D-TECH to strengthen connectivity across critical facility zones supporting its Intelligent Transportation System (ITS) permanent network. The assignment combined outside-plant fibre pathways, structured copper, passive termination, rack infrastructure, Aruba switching, identification and commissioning within one coordinated project.

1,387 m

OFC backbone

1,525 m

HDPE pathway

1,238 m

FTP cabling

Business requirement

- Create dependable, high-speed connectivity between the Wheel Test Lab, LMC Zone-4, BIAS DC and supporting operational areas.
- Introduce a documented and serviceable physical layer through disciplined termination, rack dressing, identification and tagging.
- Integrate new Aruba switching with the existing environment while limiting disruption to ongoing operations.

D-TECH VALUE PROPOSITION

A turnkey network delivery model: survey and route planning, material coordination, installation, termination, configuration, testing and documentation—managed as one accountable scope.

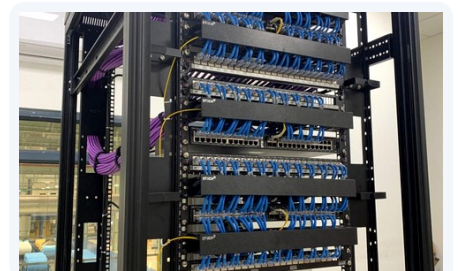
DESIGNED AS ONE COORDINATED SYSTEM

Fibre backbone, structured cabling and managed switching

The implemented scope connected multiple plant zones and combined pathway readiness, passive network build, active equipment configuration and physical-layer documentation.

Workstream	Verified delivery quantity / coverage
Optical fibre backbone	1,387 m across Wheel Test Lab, LMC Zone-4 and BIAS DC
HDPE pathway	1,525 m across approved cable routes
FTP structured cabling	1,238 m for New Engineering Store and LMC Zone-4 Cubic
Fibre termination	60 splices and 4 LIUs
Edge infrastructure	12 I/Os, 5 patch panels, 10 faceplates and one 15U rack
Network equipment	3 Aruba switch locations plus stacking, SFP and power components
Asset identification	Cables, panels, LIUs and 29 racks tagged for supportability

Implementation evidence



EXECUTION BUILT AROUND CONTINUITY AND CONTROL

A phased rollout across live plant zones

The project required work across diverse internal routes and active operating areas. D-TECH used site-led planning, phased execution and close coordination with MRF's IT and operations teams.

Challenge-to-response matrix

Delivery consideration	D-TECH response
Complex route and cable laying	Surveyed the facility, validated cable routes and sequenced OFC/HDPE work to reduce avoidable disruption.
Network continuity during change	Executed installation and switch integration in controlled phases, aligning cutovers with site stakeholders.
Multi-department coordination	Maintained working communication with IT and operations teams for access, logistics and activity planning.
Integration with installed systems	Applied Aruba switching expertise to configure, stack and connect the new equipment with the existing network.

Value embedded in the delivery model

CONTROL

Defined routes, quantities and locations created a traceable delivery baseline.

SERVICEABILITY

Labelling and rack identification improved troubleshooting readiness and asset visibility.

SCALABILITY

Fibre and managed switching established capacity for future network expansion.

All statements reflect the supplied project record; no modelled ROI or unverified KPI has been added.

FROM INSTALLATION SCOPE TO OPERATIONAL VALUE

A documented, future-ready plant network foundation

The completed project improved connectivity and created a more structured network environment for the ITS permanent network. Value came from both the installed capacity and the discipline applied to identification, configuration and handover.

CONNECTIVITY

A fibre-and-copper network spanning the identified operational zones, supported by Aruba switching.

RELIABILITY

Planned pathways, fibre termination and tested active components reduced dependence on ad-hoc links.

OPERABILITY

Tagged cables, panels, LIUs and racks support faster identification and more systematic maintenance.

FUTURE READINESS

A scalable backbone and managed-switch architecture provide a base for additional devices and services.

CUSTOMER FEEDBACK

“The professionalism, expertise and meticulous planning exhibited by the D-TECH team ensured the successful enhancement of our IT infrastructure.”

Mr. Kevin Mathew | MRF Limited

A foundation designed to be supportable, scalable and ready for the next requirement.

D-TECH Solution Integrators Pvt. Ltd. | sales@dtechindia.com | www.dtechindia.com